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HTTP

Hypertext Transfer Protocol (HTTP) is an application-layer protocol for transmitting hypermedia documents, such as HTML. It was designed for communication between web browsers and web servers, but it can also be used for other purposes. HTTP follows a classical client-server model, with a client opening a connection to make a request, then waiting until it receives a response. HTTP is a stateless protocol, meaning that the server does not keep any data (state) between two requests.

Tutorials

Learn how to use HTTP with guides and tutorials.

Overview of HTTP

The basic features of the client-server protocol: what it can do and its intended uses.

HTTP Cache

Caching is very important for fast Web sites. This article describes different methods of caching and how to use HTTP Headers to control them.

HTTP Cookies

How cookies work is defined by RFC 6265. When serving an HTTP request, a server can send a `Set-Cookie` HTTP header with the response. The client then returns the cookie's value with every request to the same server in the form of a `Cookie` request header. The cookie can also be set to expire on a certain date, or

restricted to a specific domain and path.

Cross-Origin Resource Sharing (CORS)

Cross-site HTTP requests are HTTP requests for resources from a **different domain** than the domain of the resource making the request. For instance, an HTML page from Domain A (`http://domaina.example/`) makes a request for an image on Domain B (`http://domainb.foo/image.jpg`) via the `img` element. Web pages today very commonly load cross-site resources, including CSS stylesheets, images, scripts, and other resources. CORS allows web developers to control how their site reacts to cross-site requests.

HTTP Client Hints

Client Hints are a set of response headers that a server can use to proactively request information from a client about the device, network, user, and user-agent-specific preferences. The server can then determine which resources to send, based on the information that the client chooses to provide.

Evolution of HTTP

A brief description of the changes between the early versions of HTTP, to the modern HTTP/2, the emergent HTTP/3 and beyond.

Mozilla web security guidelines

A collection of tips to help operational teams with creating secure web applications.

HTTP Messages

Describes the type and structure of the different kind of messages of HTTP/1.x and HTTP/2.

A typical HTTP session

Shows and explains the flow of a usual HTTP session.

Connection management in HTTP/1.x

Describes the three connection management models available in HTTP/1.x, their

strengths, and their weaknesses.

Reference

Browse through detailed HTTP reference documentation.

HTTP Headers

HTTP message headers are used to describe a resource, or the behavior of the server or the client. Header fields are kept in an IANA registry . IANA also maintains a registry of proposed new HTTP message headers .

HTTP Request Methods

The different operations that can be done with HTTP: `GET` , `POST` , and also less common requests like `OPTIONS` , `DELETE` , or `TRACE` .

HTTP Status Response Codes

HTTP response codes indicate whether a specific HTTP request has been successfully completed. Responses are grouped in five classes: informational responses, successful responses, redirections, client errors, and servers errors.

CSP directives

The `Content-Security-Policy` response header fields allows website administrators to control resources the user agent is allowed to load for a given page. With a few exceptions, policies mostly involve specifying server origins and script endpoints.

Tools & resources

Helpful tools and resources for understanding and debugging HTTP.

Firefox Developer Tools

Network monitor

Mozilla Observatory

A project designed to help developers, system administrators, and security professionals configure their sites safely and securely.

RedBot

Tools to check your cache-related headers.

How Browsers Work (2011)

A very comprehensive article on browser internals and request flow through HTTP protocol. A MUST-READ for any web developer.

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